

# Rank, Nullity & Systems of Linear equations :-

## Row Echelon form, singularity & Rank

### \* Table of Content

- ① Revisiting Row echelon form & row reduced echelon form
- ② Rank of a Matrix
- ③ Rank - Nullity Theorem
- ④ Rank and solution of system of linear equation
- ⑤ Problem Practice on rank and nullity.

### \* Recap : Row Echelon form & Row reduced Echelon form

→ Let's recap above concepts with an example where we have a system of linear equation.

$$\begin{array}{l} a + b + 2c = 12 \\ 3a - 3b - c = 3 \\ 2a - b + 6c = 24 \end{array} \Leftrightarrow \begin{array}{c} Ax = b \\ \begin{bmatrix} 1 & 1 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 6 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 12 \\ 3 \\ 24 \end{bmatrix} \end{array} \Leftrightarrow \begin{array}{c} \text{Augmented Matrix} \\ \left[ \begin{array}{ccc|c} 1 & 1 & 2 & 12 \\ 3 & -3 & -1 & 3 \\ 2 & -1 & 6 & 24 \end{array} \right] \end{array}$$

Systems of equationMatrix formAugmented Matrix

→ Let's reduce the augmented Matrix to row echelon form and further to row reduced echelon form.

$$\left[ \begin{array}{ccc|c} 1 & 1 & 2 & 12 \\ 3 & -3 & -1 & 3 \\ 2 & -1 & 6 & 24 \end{array} \right] \xrightarrow[r_3 = r_3 - 2r_1]{r_2 = r_2 - 3r_1} \left[ \begin{array}{ccc|c} 1 & 1 & 2 & 12 \\ 0 & -6 & -7 & -33 \\ 0 & -3 & 2 & 0 \end{array} \right] \xrightarrow{r_3 = r_3 - 0.5r_2} \downarrow$$

$$\begin{array}{c}
 R_1 = R_1 - R_2 \\
 R_2 = R_2 - \frac{7}{6}R_3
 \end{array}
 \begin{array}{c}
 \left[ \begin{array}{ccc|c}
 1 & 1 & 2 & 12 \\
 0 & 1 & \frac{7}{6} & \frac{37}{6} \\
 0 & 0 & 1 & 3
 \end{array} \right]
 \xleftarrow[
 \begin{array}{c}
 R_2 \rightarrow -\frac{1}{6}R_2 \\
 R_3 \rightarrow \frac{1}{5.5}R_3
 \end{array}
 ]{
 \begin{array}{c}
 \left[ \begin{array}{ccc|c}
 1 & 1 & 2 & 12 \\
 0 & -6 & -7 & -33 \\
 0 & 0 & 5.5 & 16.5
 \end{array} \right]
 \end{array}
 }
 \end{array}$$

Row Echelon form

$$\begin{array}{c}
 \left[ \begin{array}{ccc|c}
 1 & 0 & \frac{5}{6} & \frac{39}{6} \\
 0 & 1 & 0 & 2 \\
 0 & 0 & 1 & 3
 \end{array} \right]
 \xrightarrow{R_1 = R_1 - \frac{5}{6}R_3}
 \left[ \begin{array}{ccc|c}
 1 & 0 & 0 & 4 \\
 0 & 1 & 0 & 2 \\
 0 & 0 & 1 & 3
 \end{array} \right]$$

row Reduced Echelon form

→ Row reduced Echelon form (RREF) is the simplified form of Augmented matrix but computationally expensive.

→ Pivot provide a lot of important info regarding system of linear equation.

### \* Rank of a Matrix

→ Rank can tell many interesting information about a Matrix

#### ① Number of Linearly Independent rows or columns of Matrix

$$A = \begin{bmatrix} 3 & 2 & 5 \\ 1 & 5 & 8 \\ 2 & -3 & -3 \end{bmatrix}$$

→ # Linearly Independent rows = 2

→ # Linearly Independent columns = 2

$$\hookrightarrow c_3 = \frac{9}{13}c_1 + \frac{13}{13}c_2$$

$$\hookrightarrow r_3 = r_1 - r_2$$

→ It means out of 3 rows/columns, only 2 rows/columns are truly useful (Contains useful information)

#### ② Number of pivot elements in echelon form of a Matrix

→ For a matrix in echelon form, rank is equal to number of pivots.

$$\begin{array}{l}
 a + b + 2c = 12 \\
 3a - 3b - c = 3 \\
 2a - b + 6c = 24
 \end{array}$$

$\iff$

$$\left[ \begin{array}{ccc|c}
 1 & 1 & 2 & 12 \\
 3 & -3 & -1 & 3 \\
 2 & -1 & 6 & 24
 \end{array} \right]$$

rank = 3

$$\left[ \begin{array}{ccc|c}
 1 & 1 & 2 & 12 \\
 0 & -6 & -7 & -33 \\
 0 & 0 & 5.5 & 16.5
 \end{array} \right]$$

rank = 3

↳ For a system with 3 equations and rank=3, it tells that all equation convey some new information (no redundancy)

↳ To compute Rank, Convert a matrix to row echelon form and count number of pivot elements

### ③ Number of non-zero rows in echelon form of a Matrix

$$\begin{aligned} x+2y+z &= 0 \\ 2x+4y+2z &= 0 \\ x+2y+3z &= 0 \end{aligned}$$

System of Linear equation

$\iff$

$$\left[ \begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 4 & 2 & 0 \\ 1 & 2 & 3 & 0 \end{array} \right]$$

Augmented Matrix

$\longrightarrow$

$$\left[ \begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Row Echelon Form

$$x = t \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

↑ represents a line in xy plane

↳ Rank = no of pivots in row echelon form = 2

↳ Nullity =  $3 - 2 = 1$

Nullity = Number of free variables

↑ If no of variables =  $n$   
and rank =  $r$   
so Nullity =  $(n - r)$

\* Rank-Nullity Theorem

\* Rank-Nullity Theorem

For a matrix  $A \in \mathbb{R}^{m \times n}$

$$\text{Rank} + \text{Nullity} = n$$

← Total no of variables

↑ #pivot variables

↑ #Free Variables

Nullity tells the dimension of solution space

→ The four fundamental subspace

Example: Solve the system of equations using Gaussian Elimination

$$x_1 + 2x_2 - x_3 + 3x_4 = 0$$

$$x_1 + x_2 - x_4 = 0$$

$$\rightarrow A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 1 & 1 & 0 & -1 \end{bmatrix} \xrightarrow{R_2 = R_2 - R_1} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -1 & 1 & -4 \end{bmatrix}$$

Row echelon form

→ Pivot var =  $x_1, x_2$

→ rank = 2

free var =  $x_3, x_4$

→ nullity = 2

→ Let  $x_3 = t_1$  and  $x_4 = t_2$

→ system of eq<sup>n</sup> corresponding to row echelon form is

$$x_1 + 2x_2 - x_3 + 3x_4 = 0 \quad \text{--- ①}$$

8/15

$$-x_2 + x_3 - 4x_4 = 0 \quad \text{--- ②}$$

$$x_2 = t_1 - 4t_2$$

from eq<sup>n</sup> ①

$$x_1 = -2x_2 + x_3 - 3x_4$$

$$= -2(t_1 - 4t_2) + t_1 - 3t_2$$

$$= -2t_1 + 8t_2 + t_1 - 3t_2$$

$$x_1 = -t_1 + 5t_2$$

Solution set  $x$  corresponding to  $Ax = 0$  is

$$x = \begin{bmatrix} -t_1 + 5t_2 \\ t_1 - 4t_2 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} -t_1 \\ t_1 \\ t_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 5t_2 \\ -4t_2 \\ 0 \\ t_2 \end{bmatrix} = t_1 \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} 5 \\ -4 \\ 0 \\ 1 \end{bmatrix}$$

→ Infinite solution but only 2 LI solutions.

↑ dimension of  
Solution Space = 2

Example: Solve the system of equations using Gaussian Elimination.

$$x + y + z = 2$$

$$2x + y + 3z = 5$$

$$3x + 2y + 4z = 8$$

$$\rightarrow \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 1 & 3 & 5 \\ 3 & 2 & 4 & 8 \end{array} \right] \xrightarrow[\substack{R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1}]{\substack{R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1}} \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 1 & 2 \end{array} \right] \xrightarrow{R_3 = R_3 - R_2}$$

→ System of equations corresponding to row echelon form

$$x + y + z = 2$$

$$-y + z = 1$$

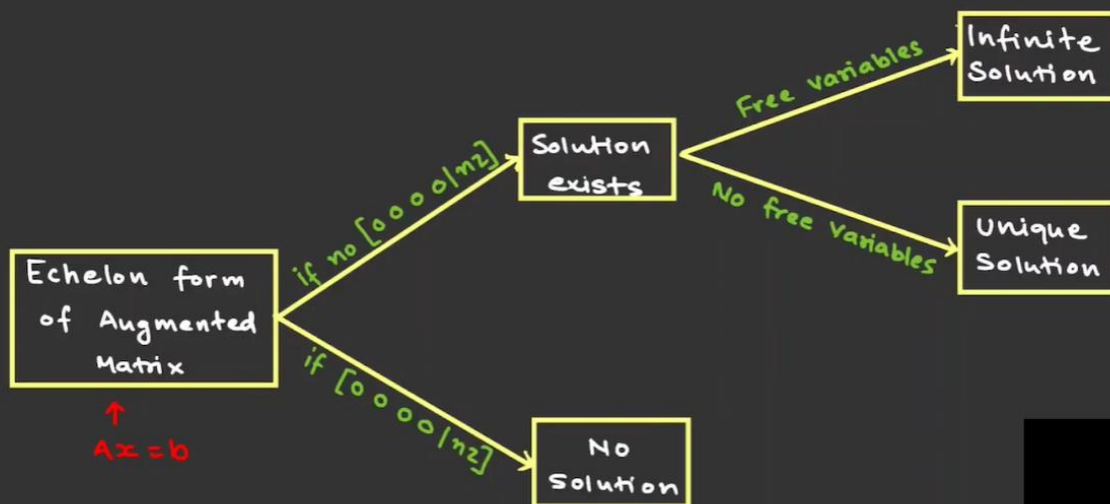
$$0x + 0y + 0z = 1 \Rightarrow 0 = 1 \leftarrow \text{Contradiction}$$

So the system of equation has no solution.

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

Row Echelon form

\* Flow chart for solution of  $Ax=b$



- Homogeneous systems ( $Ax = 0$ ) always have a solution

Example: For  $Ax=0$ , coefficient matrix  $A_{50 \times 100}$  has 20 pivot columns.

① How many Linearly Independent columns  $A$  has?

② what can you tell about Linearly Independent solutions?

→  $A_{50 \times 100} \Rightarrow$  represents a systems of Linear equation with  
→ 50 equations  
→ 100 variables

20 Pivot Variables

80 free Variable

rank = 20

20 out of 100 columns  
are Linearly Independent

nullity = 80

dimension of solution  
space = 80

solution is a 79-d  
hyperplane in  $\mathbb{R}^{80}$



## \* Rank & Solution of system of Linear equations

→ For a system of Linear equations  $Ax=b$

$A \leftarrow$  Coefficient Matrix

$[A|b] \leftarrow$  Augmented Matrix

→ Always  $\text{Rank}([A|b]) = \begin{cases} \text{Rank}(A) \Leftrightarrow \text{Solution exists} \\ 1 + \text{Rank}(A) \Leftrightarrow \text{Solution doesnot exists} \end{cases}$

↑  
Contains  $[0 \ 0 \ 0 \ 0 \ | \ n_2]$  row  
in row echelon form

→  $\text{Rank}([A|b]) = \text{Rank}(A) \Leftrightarrow b$  is linear combination of columns of  $A$

→  $\text{Rank}([A|b]) \neq \text{Rank}(A) \Leftrightarrow \text{Rank}([A|b]) = 1 + \text{Rank}(A)$   
 $\updownarrow$   
 $b$  is not linear combination of columns of  $A$

13/15

## \* Flow chart for Solution of $Ax=b$ ( $n$ variables)

